

Smartwatch for E-Bike Connectivity - A Case Study By EMotorad

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Koyal

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Market Research

Target Audience: E-bike users, fitness enthusiasts, and commuters.

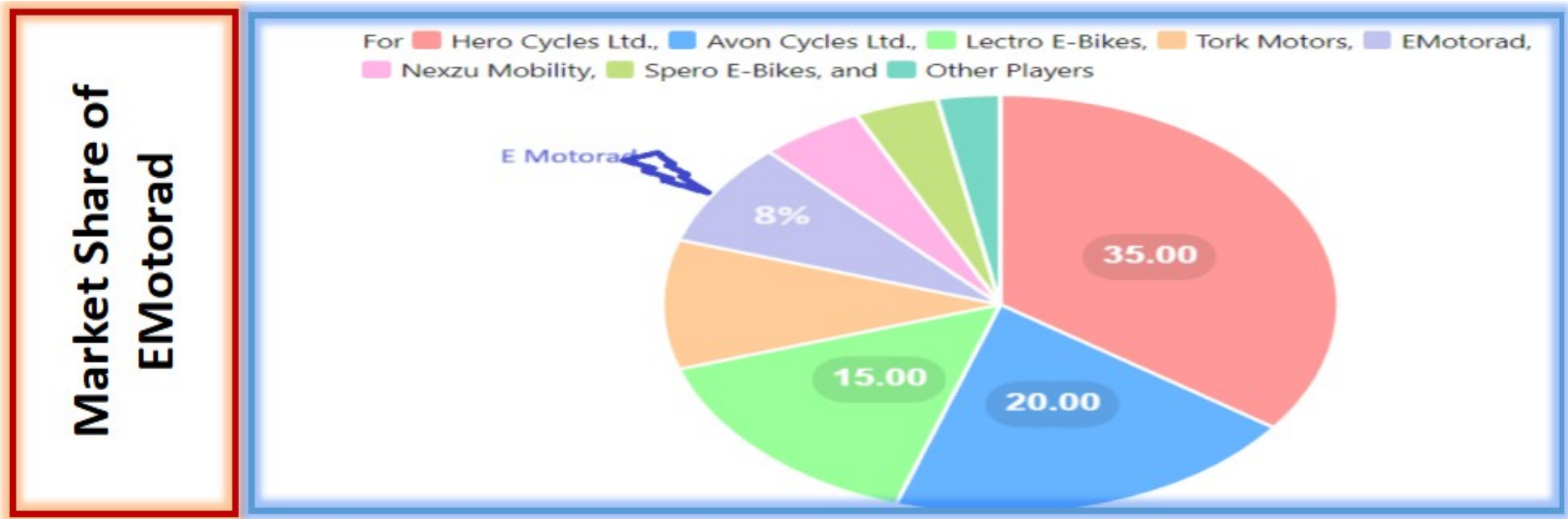
Pain Points:

- Battery monitoring
- Ride statistics
- Safety concerns

Competitive Analysis: Established smartwatch brands like Apple and Samsung offer comprehensive features but lack specific functionalities for e-bike riders.

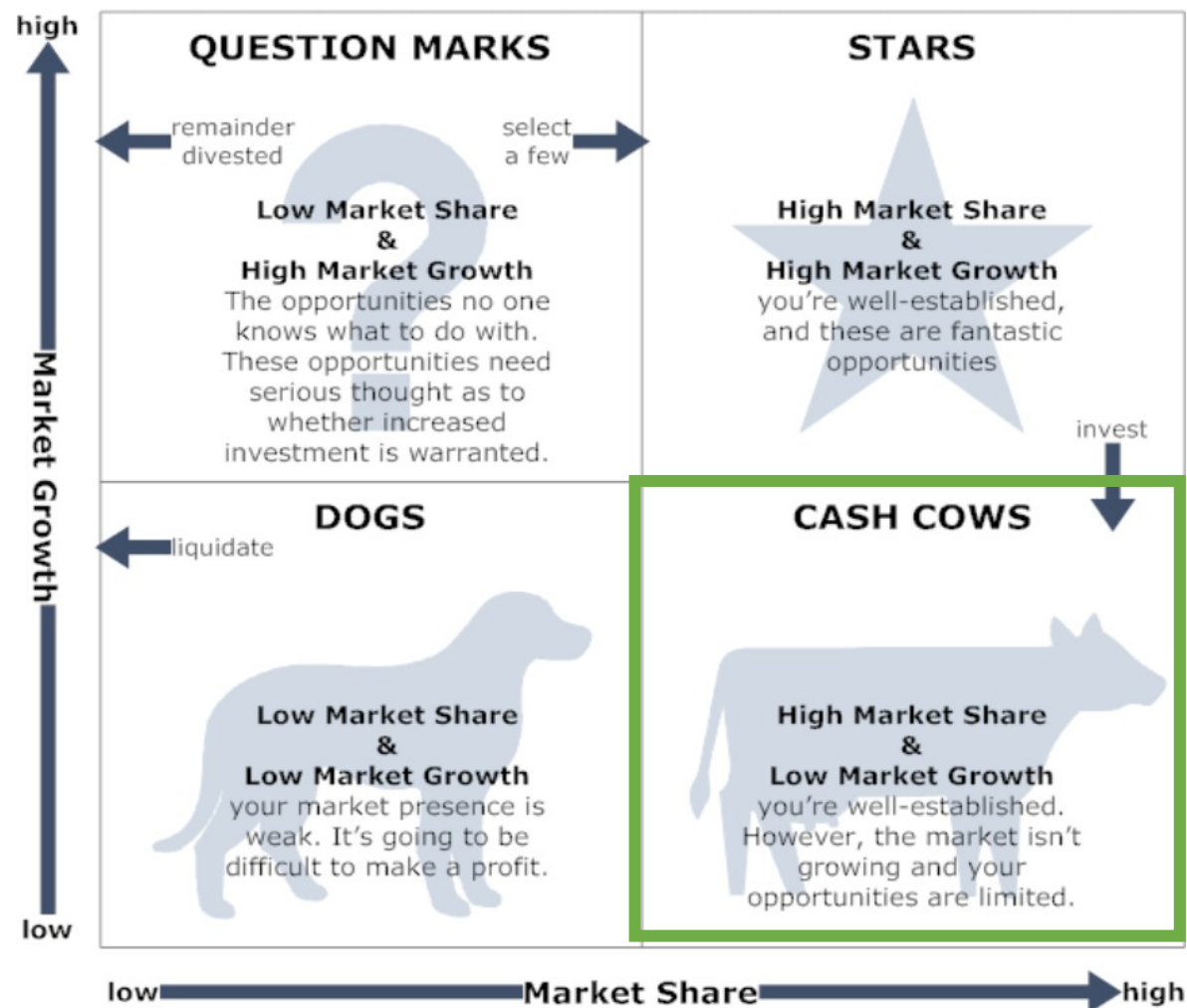
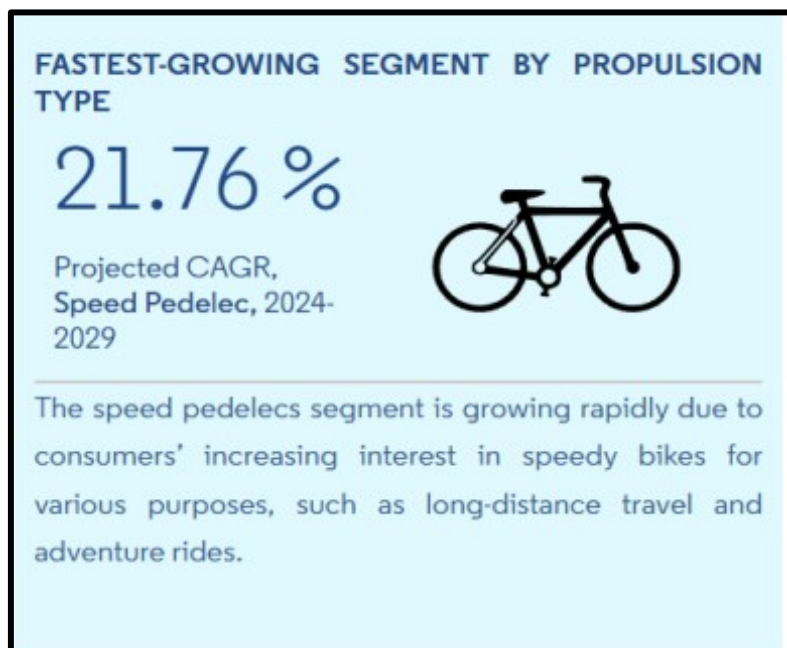
E-Bike Industry Overview

Market Value/ Year	Global (in Billion\$)	India (in Billion\$)
2021	18.58	1.02
2028	52.36	2.75



Decision

Is it worth Enough to Enter with Smartwatch in E-biCycle Industry For Emotorad?



Need To grow Market share to take best of growing Industry, through Product Development

Target Audience & its Pain Points

Demographic

Pain points

Rajiv Mehta

Office Commuter
Age: 26
IT Consultant
High Level Income
Bangalore, India

- Need for precise GPS location
- Difficulty finding nearby shops (market, chemist, etc.)
- Lack of integration with fitness apps
- Need for call alerts and music monitoring
- Fear of battery drain during commute
- Frequently forgets to charge battery

Anjali Desai

Recreational Rider
Age 29
Graphic Designer
High Middle Income
Pune, India

- Monitoring health metrics (heart rate, oxygen, calories, distance)
- Finding safe routes for leisure rides
- Tracking ride statistics hands-free
- Ensuring safety during solo rides
 - Managing music and entertainment while riding

Yoginder Singh

Town Rider
Age: 39
College Professor
Middle class
Panipat, India

- Irregular fitness routine due to busy schedule
- Difficulty monitoring e-bike battery and performance
- Safety concerns during non-crowded hours
- Worries about cycle safety in parking

Competitive Analysis Using PVP Framework

Category	Feature	Apple Watch Series 9	Garmin Forerunner 965	Samsung Galaxy Watch LTE	Garmin Edge
Price (Indian Market)		40000	67000	60000	62000
Must Have	Real-time e-bike metrics (battery, speed, range)	0	0	0	1
	Health tracking (heart rate, calories)	1	1	1	0
	GPS navigation	1	1	1	1
	Notifications (calls, messages, emails)	1	1	1	0
Performance Benefits	Safety features (fall detection, emergency alerts)	1	1	1	0
	Advanced health tracking (ECG, SpO2)	1	1	0	0
	Long battery life	0	1	1	1
	Reliable route planning	1	1	0	1
Desirable	Integration with Another devices	1	0	1	0
	Music control and integration	1	1	1	0
	Customizable watch faces	1	1	1	0
	Voice assistant integration	1	0	1	0
Total		10	9	9	4

Features BrainStroming

- GPS Location Tracking & Navigation
- Nearby Shops Finder
- Fitness Integration
- Call & Music Management
- Battery Management
- Safety & Security Features
- Ride Statistics & Health Monitoring

Features Prioritization

Feature	Reach (users/month)	Impact	Confidence (%)	Effort (person-months)	RICE Score
GPS Location Tracking & Navigation	1000	3	90	4	675
Nearby Shops Finder	800	2	85	2	680
Fitness Integration	900	2	80	2	720
Call & Music Management	700	1	95	1	665
Battery Management	1000	3	90	3	900
Safety & Security Features	700	3	80	5	336
Ride Statistics & Health Monitoring	900	2	85	4	382.5

Note: In Case of Product Redesign or Improvement, could have consider Impact/Effort Matrix

VALUE PROPOSITION

EM JivitPulse Smart Watch

- Jivit means alive or lively, combined with Pulse to reflect health monitoring and vitality.
- It incorporates elements of health, vitality, and movement with a modern and sleek design.



Technology Consideration

HARDWARE

- ▢ Heart rate and SpO2 sensors
- ▢ Bluetooth Low Energy (BLE) module
- ▢ High-resolution touch screen
- ▢ GPS module
- ▢ Long-lasting battery
- ▢ Accelerometer and gyroscope for fall detection

SOFTWARE

- ▢ Custom OS optimized for smartwatch
- ▢ Integration with popular fitness apps (Google Fit, Apple Health)
- ▢ Custom navigation and safety applications
- ▢ Music and call management interface

CONNECTIVITY

- ▢ BLE for seamless e-bike integration
- ▢ GPS for navigation
- ▢ Wi-Fi and cellular connectivity for emergency alerts and updates

Metrics

Feature Usage Metric

- ✓ GPS Navigation Usage
- ✓ Battery Management Interactions
- ✓ Nearby Shops Finder Usage
- ✓ Health and Fitness Tracking
- ✓ Safety Features Activation

Sales and Growth Metrics

- ✓ Unit Sales
- ✓ Customer Acquisition Cost (CAC)
- ✓ Lifetime Value (LTV)

User Engagement Metrics

- ✓ Daily Active Users (DAU)
- ✓ Session Length
- ✓ Retention Rate

Business Case

- Pricing: Premium pricing strategy, approximately \$299 (INR 26,000).
- Marketing: Targeted campaigns, partnerships with e-bike manufacturers like Emotorad, presence in tech and fitness expos.

Revenue Streams:

- Product Sales
- Subscription Services
- Partnerships

Potential Partnerships:

- E-bike manufacturers
- Fitness app developers
- Retail chains

THANKYOU

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